Neonatal Seizure Diagnosis and Management Guidelines

Suspected seizure in neonate <28 days CGA ⇒

attempt to suppress
after seizure control
note duration
note clinical signs
if seizure highly suspicious (>5 mins or >3 in 1 hour) or prolonged (>30 mins) or recurrent (>3 in 24 hours) or equivalent encephalopathy or more urgently suspect
discuss with neurology
consider EEG urgently

If seizure confirmed on aEEG or high clinical suspicion
First line: PHB 20 mg/kg ⇒

Second line:
(preferred) additional 10 mg/kg PHB (depending on clinical response)
or Fosphenytoin 20 mg/kg
or Phenobarbital (not well tolerated in PICU setting)
or Levetiracetam 40 mg/kg (if not already given; consider depression and/or hypotension at levels >40 mg/kg)
or Lidocaine 2 mg/kg (if not already given; consider 2 mg/kg/h x 12 in cardiac patients, CPR & patients with strong family hx of unexplained neonatal deaths; consult Neurology or Cardiology)

Third line: (whichever not used above) (20 ml/kg)
– Pro: LEV, PHT, PB, LIDO, FOS, MIDA, VPA
– Pro: LEV, pyridoxine, vit B6 (if PDE)? (We take an extra 10 mL/kg of fluid, so we can't do 20 mL/kg and growing evidence of efficacy)

Fourth line: (Midazolam etc.)
0.2 mg/kg IV bolus and start infusion at 0.2 mg/kg/hr. Assess response after 30 mins. If no response, consider further additional 0.2 mg/kg bolus and increase rate of infusion by 0.1-0.2 mg/kg/hr to a maximum of 2 mg/kg/hr

If refractory to above, or if hemodynamics will not allow for midazolam infusion, consider barbiturate infusion, + Phenobarbital (if not already given) + Lidocaine (if not already given), can drive PHB levels up to 80-100 if needed and tolerated (hypotension, requires pressor assist)

Workup for etiology of seizures ⇒
(seen late in neonatal encephalopathy, hypoglycemia, infection, bleeding diatheses,
suggested labs: CBC, CMP, Mag, sPak, coagulation studies and inflammatory markers, consider lumbar sampling if suspicious)
MRI/MRS
In correlation with neurology, consider:
non-SMA genetic/mito testing ⇒
consult Genetics, Neurology, Mitochondrial Medicine for detailed discussion of head imaging if clinical picture consistent with HIE
– Consider MRI/MRS if persistent encephalopathy
– CP if concern for infection or reduce error of interpretation
– consider serial imaging and/or advanced techniques (DWI, other sequences, MRS, CSF flow, etc.) if concerned about ischemic injury, hypoxic injury, metabolic disorder, or other genetic/structural disease
– Genetic testing: consider if there is no clear structural explanation for the encephalopathy, or if there is a family hx of early or unexplained seizures, epileptic spasms, or early encephalopathy
– Consider Genetics, Neurology, Mitochondrial Medicine, Epilepsy (PEDE), anyone (GIN), genome (+/WGL) or if brain malformations or other systemic abnormalities, consider genetic/metabolic consult

When to suspect seizure, most neonatal seizures are subtleties of clinical. Key only most often focal tonic or focal clonic ⇒

Table 1. Level of Diagnostic Evidence ⇒

Level	Definition	Course of Action
Level 1	Surface EEG or aEEG with a correlated clinical event	Treat
Level 2	Suspicious clinical with an electrically confirmed seizure pattern (aEEG or surface EEG) elsewhere in time or in other patient	Treat
Level 3	Clinical seizure either that has been treated or is recurrent	Consider level 2 workup
Level 4	Suspicious clinical event, not treated, not recurrent	Do not treat
Level 5	Non-seizure disorders treated by NICU care	Do not treat

Note: patients on ECMO may require continuous aEEG monitoring or serial conventional EEG

Prevalence at high risk for ischemic injury (HIE, stroke, cardiac arrest, sepsis, meningitis, PDA, etc.) ⇒
Clinical seizures: 1-2 per 1000
Electrical seizures: 5-10 per 1000
Electrical status: 2 per 1000

Prevalence at high risk for other breakdown with aEEG:
1. stroke
2. IVH/PVL
3. encephalopathy
4. sepsis
5. meningitis

To mitigate risk, please disconnect EEG as soon as possible, especially if there are signs of skin compromise

When to consider genetic testing or a Mitochondrial Medicine evaluation ⇒
– Think about it in all encephalopathy cases, especially those suggestive for acute or chronic metabolic/mitochondrial disease
– Fix symptoms (see chart below for dosing) ⇒
– **clinical presentation suggestive of inborn error of metabolism or mitochondrial disease** (may present with features suggesting hypoxic-ischemic encephalopathy, but with persistent encephalopathy and/or seizures and/or single etiology)
– **Family hx** suggestive of severe seizures (Dravet's, myoclonic, neonatal epilepsy of unknown etiology, epileptic spasms), neonatal encephalopathy, early infant deaths, or other neurological or metabolic disorders
– some neonates with inborn IEM-dependent epilepsy respond immediately to specific treatment, others may have a delay in response, and others may not respond at all. Consider trial of specific treatment if suspicion is high, and if no response, consider other causes. Consult Mitochondrial Medicine or Genetics if unsure about dosing or other issues
– Consider testing for causes of spina aminotrans(e), aminotrans(e)ase (AT/AS), or other neurotransmitter disorders if:
1. if unexplained or pyridoxine, pyridoxal 5-phosphate, folinic responsive seizures
2. please refer to these references to aid in depth discussion of treatment options, and consult Mitochondrial Medicine or Genetics for nuanced epilepsy care expertise (pyridoxine)

Certain causes can present with different clinical seizure types ⇒
Rapid onset seizures: Glycine, sulfite oxidase deficiency
Tonic seizures are seen rarely with hypoxic ischemic encephalopathy, but also with mitochondrial disorders, glycine encephalopathy, nonketotic dicarboxylic aciduria, sulfite oxidase deficiency, and other inborn errors of metabolism affecting the brainstem. Myoclonic seizures are seen in Dravet, Ohtahara, Lennox-Gastaut, West, Angelman, Rett syndromes, metabolic disorders such as hyperglycinemia, sulfite oxidase deficiency, and other inborn errors of metabolism and usually responsive.
Occasional metabolic seizures are focal clonic or multifocal clonic seizures that are reflexive type during or seconds and are often gerbilously distributed.
Metabolic disorders causes fibrinolysis, atelectasis, and coagulopathy and can be associated with cerebral infarction, cerebral edema, and cerebral vasospasm.
Autonomic seizures are characterized by alterations in respiration, heart rate, blood pressure, and pupil size. Seizures are observed in neonatal NE, hypoxic-ischemic encephalopathy, and intraventricular hemorrhage

Medication Reference Table

Medication	Dosage	Common Interactions	Side Effects
Phenobarbital	Loading dose: 20 mg/kg IV Maintenance: 3-5 mg/kg/day IV or PO, divided Maintenance: 2 mg/kg/day IV or PO, once daily	Respiratory depression Respiratory depression, hypotension, and death with alcohol Sedation Decreased efficacy with St. John's Wort	Drowsiness/lethargy and loss of ability (when maintenance dosing) Prolonged half life in the case of hepatic dysfunction Respiratory depression, hypotension, and death in overdose Possible paradoxical hyperkinesia
Phenytoin/ Fosphenytoin	Loading dose: 20 mg/kg IV over 30 mins Maintenance: 5 mg/kg/day IV or PO, divided Maintenance: 2 mg/kg/day IV or PO, once daily Recommend to draw pre-dose 12 hours after last dose, and 2 hours after last dose Target level 10-20 mcg/mL	Alcohol, chronic, increase Fluoxetine Fluvoxamine, acute, increase Respiratory depression, Cumar	Gingival hyperplasia Hirsutism Nystagmus, diplopia, ataxia, and other CNS effects when levels are high Hepatic dysfunction, rash Possible paradoxical hyperkinesia
Levetiracetam	Loading dose: 60 mg/kg IV Maintenance: 40 mg/kg/day IV or PO, divided Maintenance: 20 mg/kg/day IV or PO, once daily	Alcohol, acute Probenecid	Irritability and behavioral disorder Hematologic: Neutropenia, thrombocytopenia Hepatic dysfunction, rash Dose adjustment required in renal impairment
Lidocaine	Loading dose: 1 mg/kg IV Maintenance: 2 mg/kg/hr IV Consider ECG monitoring Target level 1.5-5 mcg/mL Consider ECG monitoring if dose is >2 mg/kg/hr or if patient is at risk for cardiac toxicity Consider lowering dose in patients with hepatic impairment Consult Pharmacy for assistance with "loading" and "maintenance"	Amiodarone Beta-blockers Cimetidine Fluvoxamine Mexiletine Propafenone Quinidine Verapamil	Cardiac toxicity (arrhythmia, hypotension, bradycardia, asystole) Neurologic toxicity (paresthesia, tremor, seizure, coma) Hepatic dysfunction
Midazolam	Loading dose: 0.2 mg/kg, bolus Maintenance: 0.2 mg/kg/hr, titrate to effect Maximum: 2 mg/kg/hr Consider ECG monitoring if dose is >1 mg/kg/hr	Respiratory depression Cimetidine, decreased clearance, and prolonged sedation	Respiratory depression Hypotension Seizure
Lorazepam	0.05 mg/kg, bolus, 2-3 times every 2-4 hours	Valproic acid displaces from protein binding, increasing free lorazepam and increasing sedation Valproic acid inhibits lorazepam glucuronidation, increasing half life Alcohol, chronic, increase Opioids, increase Probenecid, increase Theophylline, decrease	Sedation, ataxia, and other CNS effects Respiratory depression, hypotension, and death in overdose Paradoxical excitation Hepatic dysfunction Dose adjustment required in renal impairment
Pyridoxine	Loading dose: 100 mg IV or 100 mg, 30 minutes before lumbar puncture Maintenance: 100 mg/day IV or PO, once daily Consult Pharmacy for assistance with "loading" and "maintenance"	Isoniazid, increase Levodopa, decrease Phenobarbital, increase Phenytoin, increase Theophylline, increase Valproic acid, increase	Sensory neuropathy Ataxia Dermatitis Nausea, vomiting, and other GI effects Hepatic dysfunction Rash Hypotonia Seizure
Pyridoxal 5-phosphate	30 mg/kg/day, bolus, in 3 divided doses, for 7 days	Theophylline, increase Levodopa, decrease Phenobarbital, increase Phenytoin, increase Valproic acid, increase	Sensory neuropathy Ataxia Dermatitis Nausea, vomiting, and other GI effects Hepatic dysfunction Rash Hypotonia Seizure

Legend: PO (enteral) [accepted], IV (intravenous), IO (intraosseous), PR (rectal), SQ (subcutaneous), IM (intramuscular), Inh (inhalation), Top (topical), Trans (transdermal), Oph (ophthalmic), Otic (otic), Sub (sublingual), Buccal (buccal), Neb (nebulized), Vag (vaginal), Nas (nasal), Ung (ointment), Cream (cream), Gtt (drops), Inj (injection), Sol (solution), Susp (suspension), Tab (tablet), Cap (capsule), Supp (suppository), Pch (patch), Lozenge (lozenge), Gum (gum), Powder (powder), Aerosol (aerosol), Pen (pen injector), Pump (pump), Spray (spray), Stick (stick), Swab (swab), Wipe (wipe), Wafer (wafer), Implant (implant), Ring (ring), Sys (system), Dev (device), Pat (patch), Ins (insert), Lot (lotion), Oint (ointment), Gel (gel), Foam (foam), Paste (paste), Pow (powder